

# FACTSHEET FS06 · Same-month seasonality

**What changed on 2 October 2026.** Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS\_RERUN\_2026-10-02.md. On the corrected universe the pooled holdout spread is more clearly negative ( $t -2.21$ ): beyond 2, though not past this factsheet's 2.87 gate.

**US, 1999–2012 (pre-registered second period, run 2 October 2026).** The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net:  $t +1.16$  (2013–2026:  $-2.83$ ), reversed. Long-only alpha vs the equal-weight universe:  $+2.2\%$  a year,  $t +0.61$  (2013–2026:  $-3.17$ ), reversed. The 2013–2026 US loss did not repeat; it reads as specific to that period. Registered verdicts are unchanged. Pre-registration, method and every result: PREREG\_US\_1999\_2012.md.

## Do stocks repeat their best calendar month?

Henrik Gjernerig · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

|                                                    |                                                                                      |                                                                                              |
|----------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| STRATEGY                                           | ORIGIN                                                                               | FAMILY                                                                                       |
| Return seasonality (cross-sectional)               | Heston & Sadka (2008)                                                                | Calendar & seasonality                                                                       |
| UNIVERSES                                          | BACKTEST                                                                             | BOOKS                                                                                        |
| US · EU · UK · DK · SCANDI · World (point-in-time) | Feb 2013 – Aug 2026, monthly, net of costs                                           | Long/short and long-only (stocks with the highest past returns in the coming calendar month) |
| REBALANCE                                          | STATUS                                                                               |                                                                                              |
| Monthly                                            | Registered gate $ t  > 2.87$ over 12 cells; every cell in the programme trial ledger |                                                                                              |

**The claim.** A stock that did well relative to others in, say, March tends to do well again in March in later years, at annual lags up to 20 years. Heston & Sadka (2008) found it in US stocks 1965–2002; Keloharju, Linnainmaa & Nyberg (2016) found the same across markets and asset classes. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned  $-14.0\%$  to  $+3.4\%$  a year and was positive in 1 of the six; no universe passes the 2.87 gate (significantly negative in EU, DK). The long-only book (stocks with the highest past returns in the coming calendar month) had a higher Sharpe than the equal-weight universe in 1 of the six (UK), with alphas of  $-5.8\%$  to  $+3.0\%$ , none past the gate. **What goes wrong (§9):** the main drags are costs of about  $3.9\%$  a year, a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (appendix C):** Calendar & seasonality; its returns sit closest to the momentum cluster of the signal library. **360° view (appendix**

**D):** nearest neighbours Size, 55-day breakout; after its closest neighbours and the market it keeps an alpha of -8.2% to +3.0% (largest |t| 2.3), so it carries some information of its own.

Layout revised 2 October 2026 (verdict and scorecard, Sharpe anatomy, fit with the other strategies; detail moved to the appendix). No number changed.

## 1. Verdict and scorecard

**Verdict.** Of the 12 registered cells (two books × six universes), **0 pass** the |t| > 2.87 gate and **4 are significantly negative** (L/S EU, L/S DK, long-only US, long-only EU). Programme-wide (772 trials, |t| > 3.99): nothing survives. US 1999–2012, a period the factsheet was not built on: L/S t +1.16, long-only alpha t +0.61.

| Universe     | L/S t          | L/S Sharpe | Long-only alpha t | Long-only Sharpe vs equal-weight | Long-only max DD |
|--------------|----------------|------------|-------------------|----------------------------------|------------------|
| US           | -2.83          | -0.70      | -3.17 ✕ (neg.)    | 0.49 vs 0.85                     | -38%             |
| EU           | -3.14 ✕ (neg.) | -0.76      | -2.92 ✕ (neg.)    | 0.36 vs 0.75                     | -34%             |
| UK           | +0.79          | 0.24       | +0.99             | 0.76 vs 0.68                     | -37%             |
| DK           | -3.40 ✕ (neg.) | -0.94      | -2.30             | 0.38 vs 0.83                     | -46%             |
| SCANDI       | -2.22          | -0.63      | -1.25             | 0.65 vs 0.89                     | -38%             |
| World        | -1.07          | -0.27      | -1.04             | 0.58 vs 0.71                     | -41%             |
| US 1999–2012 | +1.16          | 0.29       | +0.61             | 0.52 vs 0.49                     | –                |

✓ passes the 2.87 gate (Bonferroni over 12 cells); ✕ significantly negative. 2013–2026 unless stated.

## 2. Headline performance

| Universe | Names / groups | L/S return / yr | L/S t | L/S CAPM alpha (t) | L/S beta | Long-only CAGR | Long-only Sharpe | Long-only max DD | EW CAGR | EW Sharpe | Size-weighted CAGR |
|----------|----------------|-----------------|-------|--------------------|----------|----------------|------------------|------------------|---------|-----------|--------------------|
| US       | 501 / 10       | -8.9%           | -2.83 | -9.1% (-2.8)       | 0.02     | +7.9%          | 0.49             | -38%             | +12.6%  | 0.85      | +14.9%             |
| EU       | 264 / 10       | -9.6%           | -3.14 | -8.1% (-2.6)       | -0.14    | +4.8%          | 0.36             | -34%             | +10.2%  | 0.75      | +10.2%             |

| Universe | Names / groups | L/S return / yr | L/S t | L/S CAPM alpha (t) | L/S beta | Long-only CAGR | Long-only Sharpe | Long-only max DD | EW CAGR | EW Sharpe | Size-weighted CAGR |
|----------|----------------|-----------------|-------|--------------------|----------|----------------|------------------|------------------|---------|-----------|--------------------|
| UK       | 324 / 10       | +3.4%           | 0.79  | +2.0% (0.5)        | 0.14     | +13.3%         | 0.76             | -37%             | +9.0%   | 0.68      | +7.8%              |
| DK       | 20 / 3         | -14.0%          | -3.40 | -13.5% (-3.2)      | -0.04    | +5.3%          | 0.38             | -46%             | +12.0%  | 0.83      | +10.2%             |
| SCANDI   | 70 / 5         | -7.8%           | -2.22 | -6.4% (-1.7)       | -0.11    | +9.5%          | 0.65             | -38%             | +12.2%  | 0.89      | +11.0%             |
| World    | 1087 / 10      | -2.6%           | -1.07 | -2.1% (-0.8)       | -0.05    | +9.6%          | 0.58             | -41%             | +10.5%  | 0.71      | +10.3%             |

Monthly, local currency (World in USD). L/S = equal-weighted stocks with the highest past returns in the coming calendar month minus stocks with the lowest, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

#### Same-month seasonality on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjernerig · Project 10 factsheets

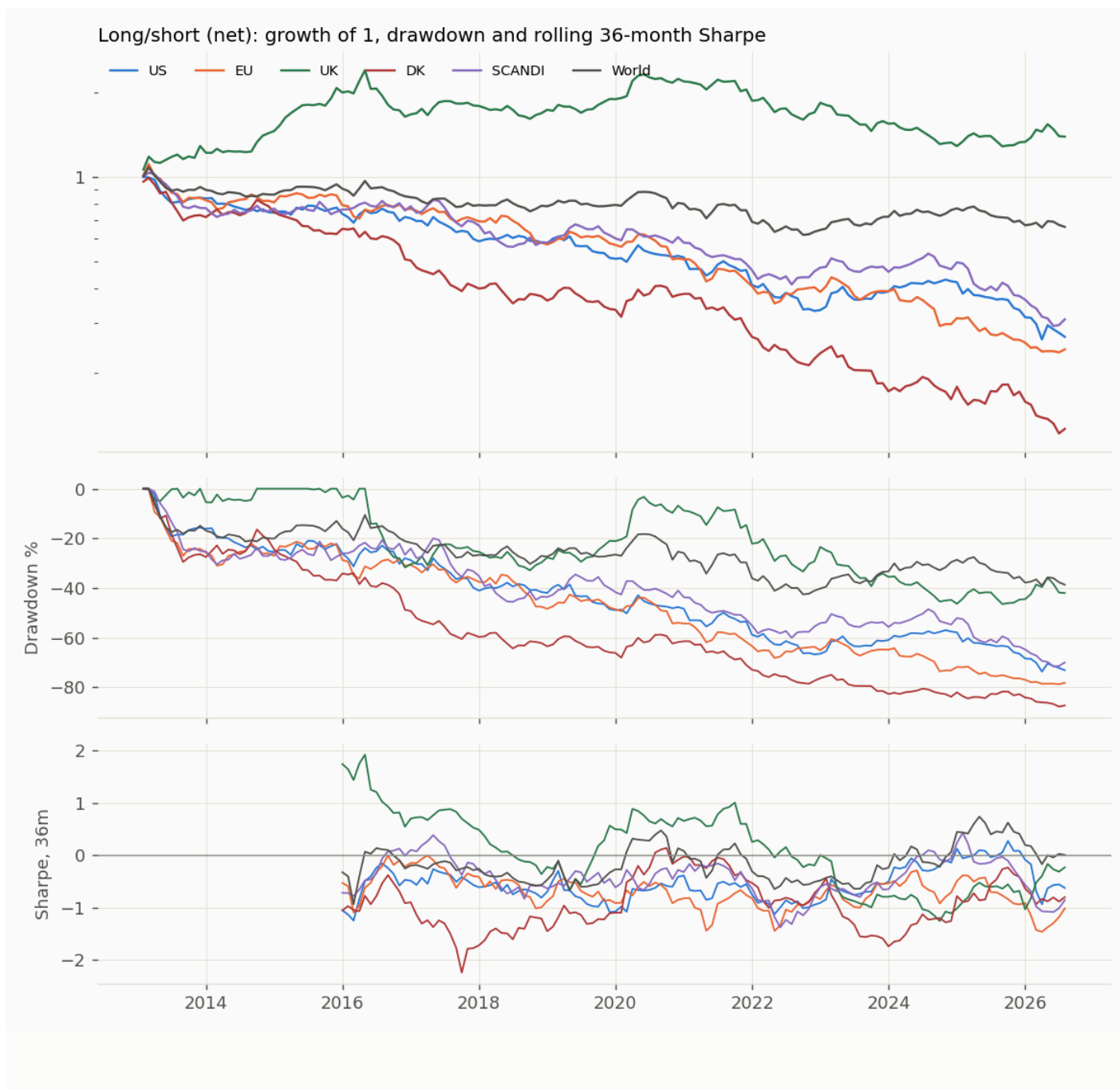
**Read with care.** A long/short book with a negative beta shows a large CAPM alpha in a rising market. The factor-model alpha in section 7, after momentum, value and the other themes, is the better guide: here the L/S CAPM alpha ranges -13.5% to +2.0% and the factor alpha -9.5% to +1.1%.

### 3. Where the Sharpe comes from

Sharpe = annual return ÷ annual volatility. The return splits into the part the market explains (beta × market return) and the rest (alpha, net of costs); the volatility into the market's share and the strategy's own. The last column is the Sharpe the book would have with its market exposure hedged out (before hedging costs).

| Universe | Book      | Return / yr | = market part | + alpha | Costs / yr (inside the return) | Volatility | of which market | Sharpe | Sharpe, market hedged |
|----------|-----------|-------------|---------------|---------|--------------------------------|------------|-----------------|--------|-----------------------|
| US       | L/S       | -8.9%       | +0.2%         | -9.1%   | -5.0%                          | 12.6%      | 0.2%            | -0.70  | -0.72                 |
| US       | Long-only | +9.6%       | +15.3%        | -5.8%   | –                              | 19.6%      | 18.1%           | 0.49   | -0.77                 |
| EU       | L/S       | -9.6%       | -1.5%         | -8.1%   | -4.9%                          | 12.7%      | 2.0%            | -0.76  | -0.65                 |
| EU       | Long-only | +6.1%       | +11.5%        | -5.4%   | –                              | 16.9%      | 15.4%           | 0.36   | -0.76                 |
| UK       | L/S       | +3.4%       | +1.4%         | +2.0%   | -4.8%                          | 13.8%      | 2.0%            | 0.24   | 0.15                  |
| UK       | Long-only | +14.3%      | +11.3%        | +3.0%   | –                              | 18.7%      | 16.6%           | 0.76   | 0.35                  |
| DK       | L/S       | -14.0%      | -0.5%         | -13.5%  | -4.0%                          | 14.9%      | 0.6%            | -0.94  | -0.91                 |
| DK       | Long-only | +6.7%       | +12.5%        | -5.8%   | –                              | 17.5%      | 15.1%           | 0.38   | -0.66                 |
| SCANDI   | L/S       | -7.8%       | -1.4%         | -6.4%   | -4.6%                          | 12.4%      | 1.6%            | -0.63  | -0.52                 |
| SCANDI   | Long-only | +10.3%      | +12.9%        | -2.5%   | –                              | 15.9%      | 14.5%           | 0.65   | -0.39                 |
| World    | L/S       | -2.6%       | -0.5%         | -2.0%   | -4.9%                          | 9.6%       | 0.7%            | -0.27  | -0.21                 |
| World    | Long-only | +10.9%      | +12.6%        | -1.7%   | –                              | 18.8%      | 17.9%           | 0.58   | -0.29                 |

Monthly, 2013–2026,  
market = equal-weight  
universe. Interactive  
version: the Strategy  
Cockpit.



## 4. How it fits next to the market and the other strategies

A long/short book improves a market portfolio when its own Sharpe beats the hurdle: its correlation with the market  $\times$  the market's Sharpe. A negative correlation makes the hurdle negative, so even a book with a small positive Sharpe diversifies; the size of the gain still depends on that Sharpe.

| Universe | Corr. with market | Avg corr. with the other strategies | Most similar strategy | Market Sharpe | Hurdle Sharpe | This L/S Sharpe | Verdict | L/S in the worst 10% of market months | Market in those months |
|----------|-------------------|-------------------------------------|-----------------------|---------------|---------------|-----------------|---------|---------------------------------------|------------------------|
| US       | +0.02             | +0.02                               | Size (-0.21)          | 0.85          | +0.02         | -0.70           | dilutes | -2.0%                                 | -7.3%                  |
| EU       | -0.16             | +0.08                               | Size (-0.29)          | 0.75          | -0.12         | -0.76           | dilutes | +0.2%                                 | -6.5%                  |
| UK       | +0.15             | +0.04                               | Size (+0.28)          | 0.68          | +0.10         | 0.24            | adds    | -0.7%                                 | -6.6%                  |

| Universe                                                                                                                                                                                            | Corr. with market | Avg corr. with the other strategies | Most similar strategy       | Market Sharpe | Hurdle Sharpe | This L/S Sharpe | Verdict | L/S in the worst 10% of market months | Market in those months |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------|-----------------------------|---------------|---------------|-----------------|---------|---------------------------------------|------------------------|
| DK                                                                                                                                                                                                  | -0.04             | -0.00                               | Residual momentum (-0.13)   | 0.83          | -0.03         | -0.94           | dilutes | -1.5%                                 | -7.2%                  |
| SCANDI                                                                                                                                                                                              | -0.13             | +0.03                               | Short-term reversal (-0.15) | 0.89          | -0.11         | -0.63           | dilutes | -0.0%                                 | -6.9%                  |
| World                                                                                                                                                                                               | -0.08             | +0.04                               | Size (-0.31)                | 0.71          | -0.05         | -0.27           | dilutes | -0.6%                                 | -7.5%                  |
| <i>Long/short (net), monthly, 2013–2026; market = equal-weight universe; other strategies = the long/short books of FS01–FS12. The Strategy Cockpit lets you build books of several strategies.</i> |                   |                                     |                             |               |               |                 |         |                                       |                        |

## 5. Strategy description

A stock that did well relative to others in, say, March tends to do well again in March in later years, at annual lags up to 20 years. Heston & Sadka (2008) found it in US stocks 1965–2002; Keloharju, Linnainmaa & Nyberg (2016) found the same across markets and asset classes.

**Why it might work.** Recurring demand and information flows tied to the calendar (earnings seasons, dividend timing, fund flows); Keloharju et al. (2016) argue it reflects seasonal variation in risk premia.

| Rule      | Original (Heston & Sadka (2008))                                   | This factsheet                                                                                          |
|-----------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Signal    | Average return in the same calendar month over the past 1–20 years | Average return in the coming calendar month over all prior years available (1 year in 2013, 13 by 2026) |
| Portfolio | Decile L/S, one month                                              | Extreme quantile L/S, one month                                                                         |
| Weighting | Equal weighted                                                     | Equal weighted                                                                                          |

## 6. Method: data, signal and portfolio

Data load

| Universe | Prices                                                                  | Membership                                                                                                                                                                                                                | Names eligible (median) | Currency                   |
|----------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|
| US       | Sharadar SEP closeadj (total return), 805 tickers incl. delisted        | <b>Point-in-time:</b> the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)                                                                                                           | ~503                    | USD                        |
| EU       | Project1 yfinance cache, Adj Close (TR)                                 | <b>Point-in-time:</b> union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced | ~238                    | local (EUR, SEK, DKK, PLN) |
| UK       | Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×) | <b>Point-in-time:</b> FTSE 100 + FTSE 250, quarterly snapshots; 75% priced                                                                                                                                                | ~242                    | GBP                        |
| DK       | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMX Copenhagen 25, quarterly snapshots; 91% priced                                                                                                                                                  | ~19                     | DKK                        |
| SCANDI   | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced                                                                                                                                 | ~62                     | local (DKK, SEK, EUR)      |
| World    | US Sharadar + UK + EU above                                             | <b>Point-in-time</b> union of the three; local-currency returns summed, not converted                                                                                                                                     | ~963                    | mixed                      |

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Price history starts in 2011–2012, so the signal averages 1 prior year in 2013 and at most 13 by 2026; Heston & Sadka use up to 20. The early years test a much noisier signal than the paper's.

## Signal creation

$SEAS_{i,t}$  = mean of  $r_{i,m}$  over the months  $m$  with the same calendar month as  $t+1$  in earlier years. Signals use only information up to month-end  $t$  and are traded in month  $t+1$ . Eligibility: in the point-in-time universe on any of the last five trading days of month  $t$ .

## Model build

- Sort eligible stocks into equal-count groups (deciles  $\geq 100$  names, quintiles 50–99, terciles  $< 50$ ; fixed per universe).
- **Long/short** = stocks with the highest past returns in the coming calendar month minus stocks with the lowest; **long-only** = stocks with the highest past returns in the coming calendar month; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash

financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

## 7. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West  $t$ , 6 lags.

### Long/short

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                                          |
|----------|------------------------|------------|----------|----------------|------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | -8.3%      | -2.84    | 0.29           | mkt +0.18 (+2.7), value -0.78 (-5.1), momentum -0.34 (-2.6), low_risk +0.37 (+2.2) |
| EU       | French Europe 5F + WML | -7.9%      | -2.40    | 0.15           | HML -0.31 (-2.1)                                                                   |
| UK       | JKP GBR 7 themes       | +0.7%      | 0.20     | 0.22           | mkt +0.19 (+2.9), quality +0.80 (+2.8), short_term_reversal -0.53 (-3.0)           |
| DK       | JKP DNK 7 themes       | -9.5%      | -2.60    | 0.06           | none  t  ≥ 2                                                                       |
| SCANDI   | French Europe 5F + WML | -4.9%      | -1.45    | 0.09           | HML -0.34 (-2.2)                                                                   |
| World    | JKP World 7 themes     | +1.1%      | 0.47     | 0.30           | mkt +0.12 (+2.6), value -0.76 (-5.4), momentum -0.33 (-3.2), low_risk +0.38 (+2.9) |

### Long-only

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                      |
|----------|------------------------|------------|----------|----------------|----------------------------------------------------------------|
| US       | JKP USA 7 themes       | -2.3%      | -1.31    | 0.90           | mkt +1.10 (+24.2), size -0.31 (-2.7), momentum -0.18 (-2.2)    |
| EU       | French Europe 5F + WML | +1.5%      | 0.49     | 0.68           | Mkt-RF +0.80 (+12.3), CMA -0.47 (-2.5)                         |
| UK       | JKP GBR 7 themes       | +7.4%      | 2.61     | 0.74           | mkt +0.70 (+13.3), low_risk -0.82 (-3.9), quality +0.86 (+3.5) |
| DK       | JKP DNK 7 themes       | +1.8%      | 0.62     | 0.60           | mkt +0.61 (+8.0), low_risk -0.57 (-3.9), quality +0.23 (+2.2)  |
| SCANDI   | French Europe 5F + WML | +6.8%      | 2.25     | 0.58           | Mkt-RF +0.69 (+10.8), WML -0.19 (-2.7)                         |
| World    | JKP World 7 themes     | +4.8%      | 1.97     | 0.82           | mkt +0.89 (+13.9), momentum -0.22 (-2.4)                       |

## 8. Statistical detail

- **Gate:**  $|t| > 2.87$  (2 books  $\times$  6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 1 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -13.5% to +2.0%,  $|t|$  past the gate in 1 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 1 of the six; alpha -5.8% to +3.0%, passing in 0 of 6.

- **Probabilistic Sharpe** of the L/S,  $P(\text{true Sharpe} > 0)$ : US 0.01, EU 0.00, UK 0.82, DK 0.00, SCANDI 0.01, World 0.16.
- **Publication decay**: gross L/S -10.8% to +15.0% a year in 2013–19 against -9.2% to +2.8% in 2020–26.

## 9. What goes wrong, and how to fix it

**Method.** Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate  $p < 0.0020$ .

### 9.1 Diagnosis (design window 2013–2019)

| Universe | L/S gross / yr | Beta  | Beta drag / yr | CAPM alpha (gross) | Costs / yr | Long leg vs EW / yr | EW vs short leg / yr | Turnover / month | Worst months                           |
|----------|----------------|-------|----------------|--------------------|------------|---------------------|----------------------|------------------|----------------------------------------|
| US       | -4.0%          | -0.13 | -1.7%          | -2.3%              | -4.1%      | -3.2%               | -0.8%                | 172%             | 2013-05 -9%, 2019-06 -7%, 2016-11 -7%  |
| EU       | -2.2%          | -0.11 | -1.2%          | -1.1%              | -4.0%      | -2.4%               | +0.1%                | 167%             | 2013-04 -9%, 2016-03 -8%, 2016-01 -8%  |
| UK       | +15.0%         | 0.19  | +2.3%          | +12.7%             | -4.0%      | +10.5%              | +4.5%                | 166%             | 2016-06 -14%, 2016-09 -7%, 2016-08 -6% |
| DK       | -10.8%         | -0.17 | -2.6%          | -8.2%              | -3.2%      | -4.0%               | -6.8%                | 135%             | 2016-12 -11%, 2013-07 -8%, 2013-09 -7% |
| SCANDI   | -1.6%          | -0.30 | -4.6%          | +3.0%              | -3.8%      | +0.4%               | -2.0%                | 158%             | 2013-09 -8%, 2017-08 -7%, 2018-02 -6%  |
| World    | +1.9%          | 0.00  | +0.0%          | +1.8%              | -4.1%      | +1.0%               | +0.9%                | 169%             | 2013-04 -6%, 2016-06 -6%, 2013-05 -5%  |

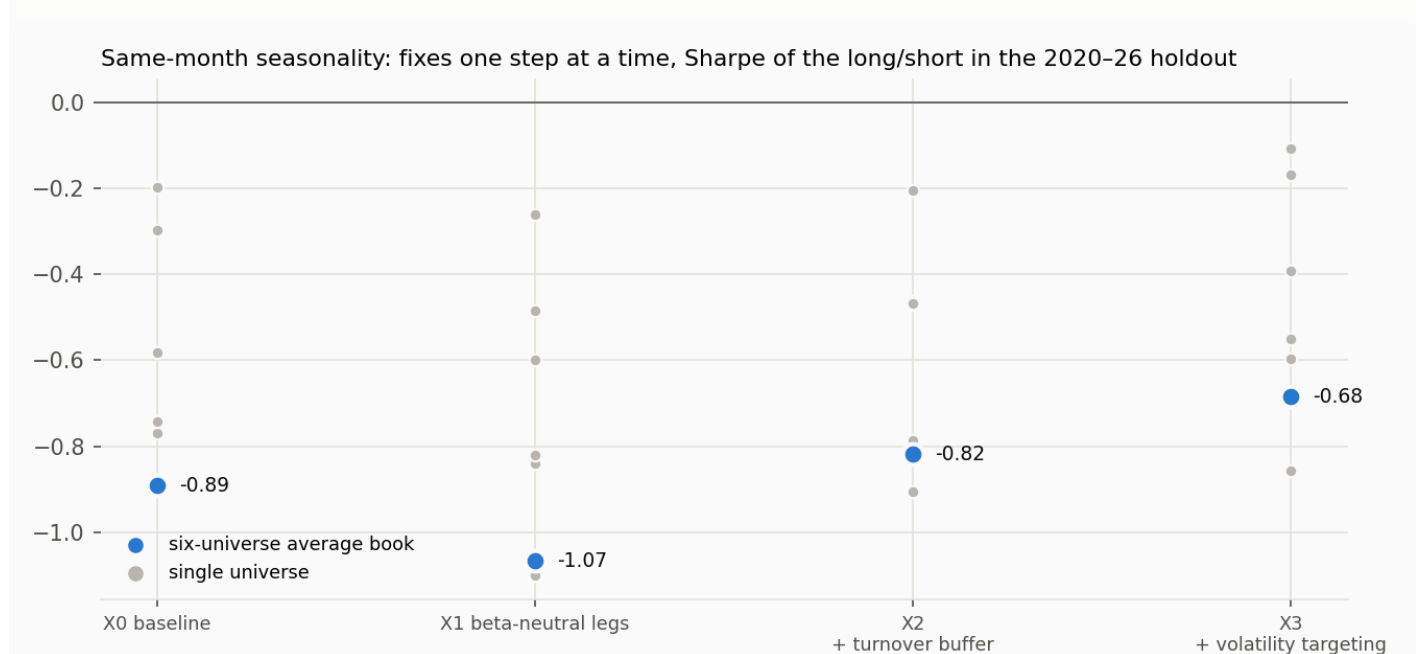
On average across the six universes the gross spread was -0.3% a year, of which the market exposure (beta -0.09) contributed -1.3%; the beta-adjusted spread (CAPM alpha) was +1.0%. Costs took 3.9% a year at 161% monthly turnover across both legs. The long leg beat the universe by +0.4% and the short leg lagged it by -0.7% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages -1.7%, +0.9%, -0.6% a year.

### 9.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

| Step                      | Design Sharpe (avg) | $\Delta$ design (p) | Holdout Sharpe (book) | $\Delta$ holdout (p) | Universes improved | Holdout return / yr | Holdout max DD | Holdout alpha vs EW (t) | Costs / yr |
|---------------------------|---------------------|---------------------|-----------------------|----------------------|--------------------|---------------------|----------------|-------------------------|------------|
| X0 baseline               | -0.47               | –                   | -0.89                 | –                    | –                  | -8.2%               | -50%           | -8.3% (-2.1)            | 3.9%       |
| X1 beta-neutral legs      | -0.46               | -0.05 (p 0.705)     | -1.07                 | -0.18 (p 0.840)      | 0 of 6             | -9.2%               | -53%           | -9.2% (-2.6)            | 3.7%       |
| X2 + turnover buffer      | -0.46               | +0.02 (p 0.406)     | -0.82                 | +0.25 (p 0.007)      | 6 of 6             | -6.6%               | -45%           | -6.3% (-1.7)            | 2.9%       |
| X3 + volatility targeting | -0.32               | +0.18 (p 0.323)     | -0.68                 | +0.13 (p 0.110)      | 5 of 6             | -6.3%               | -48%           | -5.9% (-1.3)            | 2.9%       |

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



- **X1 beta-neutral legs:** -0.05 design, -0.18 holdout; hurts in both windows.
- **X2 + turnover buffer:** +0.02 design, +0.25 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** +0.18 design, +0.13 holdout; helps in both windows but does not pass the gate.

**Where it ends:** holdout Sharpe -0.89 → -0.68, return -8.2% → -6.3% a year (six-universe book).

## 10. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** Price history starts in 2011–2012, so the signal averages 1 prior year in 2013 and at most 13 by 2026; Heston & Sadka use up to 20. The early years test a much noisier signal than the paper's.
5. **Trials.** Every gated cell is in the factsheet trial ledger; the fix ladder is counted inside FS03–FS12 (25 trials).

## 11. Academic references

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## 12. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse\_xs.py FS06 → build\_xs.py FS06.  
Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.

# Appendix

## A. Performance in detail

US (S&P 500 members, point-in-time), median 501 names, 10 groups

| Book                                                                                  | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]         | Sortino | Max<br>DD | Hit | t(mean) | Beta | Alpha/yr<br>(t) | CVaR 95%<br>(month) |
|---------------------------------------------------------------------------------------|--------------|--------|-------|----------------------------|---------|-----------|-----|---------|------|-----------------|---------------------|
| L/S (net)                                                                             | -8.9%        | -9.2%  | 12.6% | -0.70<br>[-1.18,<br>-0.22] | -0.91   | -74%      | 37% | -2.83   | 0.02 | -9.1%<br>(-2.8) | 8.8%                |
| Long-only (net)                                                                       | +9.6%        | +7.9%  | 19.6% | 0.49<br>[0.09,<br>0.98]    | 0.61    | -38%      | 60% | 2.43    | 1.17 | -5.8%<br>(-3.2) | 12.5%               |
| Stocks with the<br>highest past returns<br>in the coming<br>calendar month<br>(gross) | +11.6%       | +10.1% | 19.6% | 0.59<br>[0.19,<br>1.11]    | 0.79    | -36%      | 61% | 2.95    | 1.17 | -3.7%<br>(-2.0) | 12.3%               |
| Stocks with the<br>lowest (gross)                                                     | +15.4%       | +14.4% | 19.5% | 0.79<br>[0.34,<br>1.36]    | 1.20    | -33%      | 62% | 3.84    | 1.15 | +0.3%<br>(0.1)  | 11.4%               |
| EW universe                                                                           | +13.1%       | +12.6% | 15.5% | 0.85<br>[0.41,<br>1.42]    | 1.29    | -28%      | 67% | 4.20    | 1.00 | –               | 9.2%                |
| Size-weighted<br>universe                                                             | +14.9%       | +14.9% | 14.2% | 1.05<br>[0.60,<br>1.61]    | 1.73    | -24%      | 69% | 5.29    | 0.86 | +3.7%<br>(2.4)  | 8.3%                |

EU (11 national blue-chip indices, point-in-time), median 264 names, 10 groups

| Book                                                                                  | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]         | Sortino | Max<br>DD | Hit | t(mean) | Beta  | Alpha/yr<br>(t) | CVaR 95%<br>(month) |
|---------------------------------------------------------------------------------------|--------------|--------|-------|----------------------------|---------|-----------|-----|---------|-------|-----------------|---------------------|
| L/S (net)                                                                             | -9.6%        | -9.9%  | 12.7% | -0.76<br>[-1.22,<br>-0.31] | -0.93   | -79%      | 40% | -3.14   | -0.14 | -8.1%<br>(-2.6) | 9.0%                |
| Long-only (net)                                                                       | +6.1%        | +4.8%  | 16.9% | 0.36<br>[-0.12,<br>0.89]   | 0.43    | -34%      | 55% | 1.42    | 1.07  | -5.4%<br>(-2.9) | 10.2%               |
| Stocks with the<br>highest past returns<br>in the coming<br>calendar month<br>(gross) | +8.2%        | +7.0%  | 16.9% | 0.48<br>[-0.00,<br>1.02]   | 0.64    | -32%      | 56% | 1.90    | 1.07  | -3.3%<br>(-1.8) | 10.0%               |
| Stocks with the<br>lowest (gross)                                                     | +12.8%       | +11.6% | 19.5% | 0.66<br>[0.19,<br>1.19]    | 0.96    | -30%      | 61% | 2.72    | 1.21  | -0.2%<br>(-0.1) | 10.8%               |

| Book                   | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]      | Sortino | Max<br>DD | Hit | t(mean) | Beta | Alpha/yr<br>(t) | CVaR 95%<br>(month) |
|------------------------|--------------|--------|-------|-------------------------|---------|-----------|-----|---------|------|-----------------|---------------------|
| EW universe            | +10.7%       | +10.2% | 14.3% | 0.75<br>[0.29,<br>1.30] | 1.14    | -26%      | 61% | 3.19    | 1.00 | –               | 8.4%                |
| Size-weighted universe | +10.6%       | +10.2% | 13.7% | 0.77<br>[0.32,<br>1.29] | 1.21    | -25%      | 58% | 3.37    | 0.92 | +0.8%<br>(0.6)  | 7.9%                |

### UK (FTSE 350, point-in-time), median 324 names, 10 groups

| Book                                                                      | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]       | Sortino | Max<br>DD | Hit | t(mean) | Beta | Alpha/yr<br>(t) | CVaR 95%<br>(month) |
|---------------------------------------------------------------------------|--------------|--------|-------|--------------------------|---------|-----------|-----|---------|------|-----------------|---------------------|
| L/S (net)                                                                 | +3.4%        | +2.4%  | 13.8% | 0.24<br>[-0.34,<br>0.86] | 0.28    | -46%      | 50% | 0.79    | 0.14 | +2.0%<br>(0.5)  | 7.5%                |
| Long-only (net)                                                           | +14.3%       | +13.3% | 18.7% | 0.76<br>[0.25,<br>1.34]  | 1.22    | -37%      | 60% | 3.16    | 1.17 | +3.0%<br>(1.0)  | 10.3%               |
| Stocks with the highest past returns in the coming calendar month (gross) | +16.3%       | +15.6% | 18.7% | 0.87<br>[0.35,<br>1.46]  | 1.47    | -36%      | 61% | 3.60    | 1.17 | +5.0%<br>(1.7)  | 10.2%               |
| Stocks with the lowest (gross)                                            | +8.2%        | +7.0%  | 16.6% | 0.49<br>[0.02,<br>1.02]  | 0.64    | -34%      | 59% | 1.93    | 1.03 | -1.7%<br>(-0.7) | 10.3%               |
| EW universe                                                               | +9.6%        | +9.0%  | 14.2% | 0.68<br>[0.20,<br>1.28]  | 0.96    | -30%      | 58% | 2.88    | 1.00 | –               | 8.8%                |
| Size-weighted universe                                                    | +8.4%        | +7.8%  | 12.9% | 0.65<br>[0.16,<br>1.25]  | 0.90    | -28%      | 62% | 2.68    | 0.84 | +0.3%<br>(0.2)  | 7.8%                |

### DK (OMXC25, point-in-time), median 20 names, 3 groups

| Book                                                                      | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]         | Sortino | Max<br>DD | Hit | t(mean) | Beta  | Alpha/yr<br>(t)  | CVaR 95%<br>(month) |
|---------------------------------------------------------------------------|--------------|--------|-------|----------------------------|---------|-----------|-----|---------|-------|------------------|---------------------|
| L/S (net)                                                                 | -14.0%       | -14.1% | 14.9% | -0.94<br>[-1.52,<br>-0.41] | -1.12   | -88%      | 37% | -3.40   | -0.04 | -13.5%<br>(-3.2) | 9.0%                |
| Long-only (net)                                                           | +6.7%        | +5.3%  | 17.5% | 0.38<br>[-0.13,<br>0.96]   | 0.45    | -46%      | 55% | 1.40    | 0.99  | -5.8%<br>(-2.3)  | 10.6%               |
| Stocks with the highest past returns in the coming calendar month (gross) | +8.2%        | +6.9%  | 17.5% | 0.47<br>[-0.04,<br>1.06]   | 0.60    | -45%      | 56% | 1.73    | 1.00  | -4.3%<br>(-1.7)  | 10.4%               |



| Book                                                                      | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]   | Sortino | Max<br>DD | Hit | t(mean) | Beta | Alpha/yr<br>(t) | CVaR 95%<br>(month) |
|---------------------------------------------------------------------------|--------------|--------|-------|----------------------|---------|-----------|-----|---------|------|-----------------|---------------------|
| Stocks with the highest past returns in the coming calendar month (gross) | +13.0%       | +11.8% | 18.8% | 0.69<br>[0.21, 1.27] | 0.98    | -40%      | 61% | 2.90    | 1.12 | +0.4%<br>(0.2)  | 11.6%               |
| Stocks with the lowest (gross)                                            | +10.6%       | +9.1%  | 19.5% | 0.55<br>[0.08, 1.10] | 0.72    | -37%      | 59% | 2.26    | 1.16 | -2.5%<br>(-1.9) | 11.7%               |
| EW universe                                                               | +11.3%       | +10.5% | 16.0% | 0.71<br>[0.25, 1.28] | 1.02    | -30%      | 63% | 3.07    | 1.00 | –               | 9.7%                |
| Size-weighted universe                                                    | +11.0%       | +10.3% | 14.8% | 0.74<br>[0.28, 1.31] | 1.10    | -28%      | 61% | 3.19    | 0.91 | +0.7%<br>(0.8)  | 8.9%                |

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | -15.9% | -15.9% | +28.7% | -27.1% | -22.7% | -8.1%  |
| 2014 | -11.2% | -2.4%  | +11.0% | +7.7%  | -2.7%  | -6.0%  |
| 2015 | +3.9%  | +4.5%  | +45.0% | -20.5% | -0.4%  | +8.7%  |
| 2016 | -7.4%  | -10.7% | -21.0% | -18.7% | +1.1%  | -9.4%  |
| 2017 | -16.5% | -7.8%  | +9.7%  | -19.6% | -9.0%  | -7.5%  |
| 2018 | -2.9%  | -17.5% | -6.2%  | -8.1%  | -15.3% | -0.0%  |
| 2019 | -11.9% | -0.6%  | +12.4% | -9.9%  | +5.9%  | +0.2%  |
| 2020 | +1.7%  | -11.7% | +17.9% | +12.2% | -1.5%  | +2.1%  |
| 2021 | -11.0% | -16.9% | -8.4%  | -24.1% | -19.7% | -5.8%  |
| 2022 | -28.5% | -5.0%  | -17.1% | -21.9% | -5.9%  | -16.1% |
| 2023 | +16.1% | -2.9%  | -7.0%  | -18.3% | +4.0%  | +12.9% |
| 2024 | +9.9%  | -23.9% | -16.1% | -12.3% | -2.1%  | +5.9%  |
| 2025 | -21.0% | -12.0% | +0.2%  | +6.4%  | -19.7% | -8.3%  |
| 2026 | -19.8% | -7.6%  | +4.8%  | -26.4% | -17.5% | -5.0%  |

Calendar-year returns, long-only (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | +17.6% | +16.0% | +39.2% | +21.8% | +10.8% | +22.4% |
| 2014 | +8.9%  | +1.3%  | +13.7% | +17.4% | +12.2% | +0.1%  |
| 2015 | +0.2%  | +2.5%  | +35.0% | +11.4% | +13.3% | +4.1%  |

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2016 | +9.9%  | +7.0%  | +14.1% | +6.4%  | +31.3% | +3.6%  |
| 2017 | +5.5%  | +14.6% | +37.1% | +10.5% | +12.0% | +25.4% |
| 2018 | -11.5% | -24.4% | -14.3% | -12.2% | -14.7% | -13.2% |
| 2019 | +21.3% | +27.1% | +30.1% | +12.7% | +38.3% | +27.8% |
| 2020 | +11.1% | +1.6%  | +23.0% | +48.1% | +27.8% | +13.4% |
| 2021 | +20.8% | +0.3%  | +6.6%  | +4.0%  | +7.5%  | +11.1% |
| 2022 | -27.8% | -17.2% | -25.6% | -30.5% | -20.7% | -28.1% |
| 2023 | +26.6% | +15.5% | +10.7% | -1.4%  | +8.9%  | +27.4% |
| 2024 | +19.0% | +1.1%  | -5.1%  | -5.0%  | +3.1%  | +14.7% |
| 2025 | +4.0%  | +19.5% | +12.5% | +17.9% | +9.1%  | +17.8% |
| 2026 | +15.7% | +13.3% | +27.3% | -8.4%  | +4.9%  | +20.7% |

## B. Trading record and current book

The rebalance log per universe is in `results/xs_FS06_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

| Universe | Rebalances | Names long / short | Turnover long leg / month | Turnover short leg / month | L/S months positive | Worst L/S month | When    |
|----------|------------|--------------------|---------------------------|----------------------------|---------------------|-----------------|---------|
| US       | 163        | 51 / 50            | 85%                       | 87%                        | 37%                 | -11.9%          | 2026-04 |
| EU       | 163        | 27 / 26            | 86%                       | 82%                        | 40%                 | -12.9%          | 2024-10 |
| UK       | 163        | 33 / 32            | 85%                       | 82%                        | 50%                 | -14.2%          | 2016-06 |
| DK       | 163        | 7 / 6              | 64%                       | 72%                        | 37%                 | -11.1%          | 2016-12 |
| SCANDI   | 163        | 14 / 14            | 79%                       | 80%                        | 40%                 | -11.6%          | 2025-04 |
| World    | 163        | 109 / 108          | 85%                       | 84%                        | 47%                 | -9.1%           | 2022-01 |

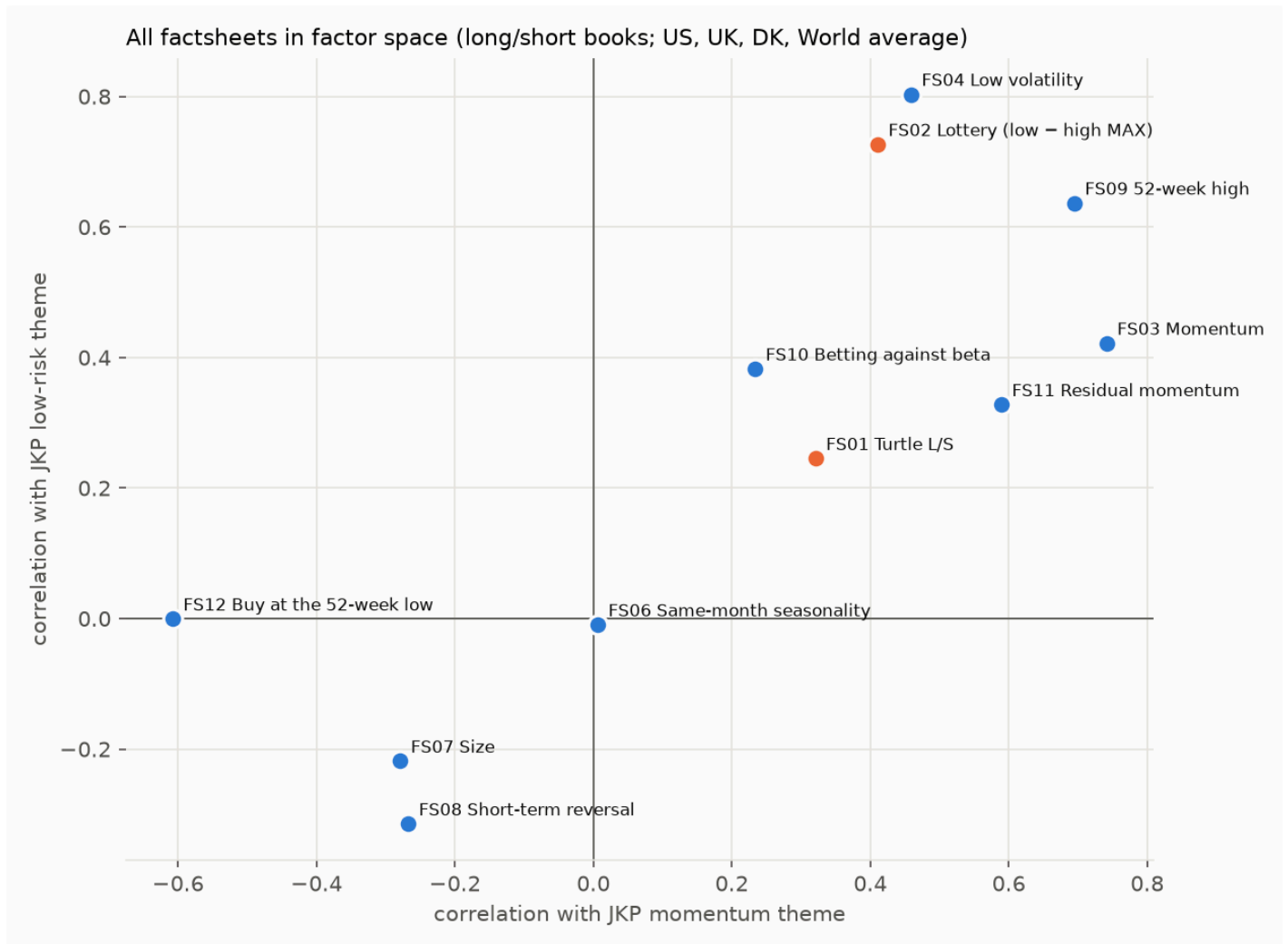
**Current book** (signal at 31 August 2026; first 8 names per leg)

| Universe | Long: stocks with the highest past returns in the coming calendar month | Short: stocks with the lowest                                           |
|----------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| US       | SNDK, GEV, APP, CEG, HOOD, PLTR, TKO, MU                                | CVNA, KVUE, DXCM, VTRS, EFX, SWKS, XYZ, INCY                            |
| EU       | ZEAL.CO, JSW.WA, HM-B.ST, NKT.CO, BOL.ST, FORTUM.HE, BKT.MC, ITX.MC     | PUIG.MC, DSFIR.AS, DTG.DE, BMPS.MI, BAVA.CO, ANE.MC, ORSTED.CO, EDEN.PA |
| UK       | CWR.L, ALFA.L, GDWN.L, TRST.L, HWG.L, JD.L, AVON.L, EWG.L               | MTLN.L, THG.L, OCDO.L, AML.L, MTRO.L, OXIG.L, DOCS.L, BCG.L             |
|          |                                                                         |                                                                         |

| Universe | Long: stocks with the highest past returns in the coming calendar month    | Short: stocks with the lowest                                                 |
|----------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| DK       | ZEAL.CO, NKT.CO, NDA-DK.CO, JYSK.CO, GMAB.CO, TRYG.CO, VWS.CO, DSV.CO      | BAVA.CO, ORSTED.CO, GN.CO, DEMANT.CO, AMBU-B.CO, ISS.CO, COLO-B.CO, NOVO-B.CO |
| SCANDI   | ZEAL.CO, HM-B.ST, NKT.CO, BOL.ST, FORTUM.HE, SWED-A.ST, NOKIA.HE, SKA-B.ST | BAVA.CO, ORSTED.CO, METSO.HE, QTCOM.HE, EQT.ST, WRT1V.HE, GN.CO, DEMANT.CO    |

## C. Classification: where does it fit?

| Dimension                         | Same-month seasonality                                                                                                                                                                    |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Series family                     | <b>Calendar &amp; seasonality</b>                                                                                                                                                         |
| Academic style                    | Return seasonality (cross-sectional)                                                                                                                                                      |
| Signal                            | Price only (daily total returns)                                                                                                                                                          |
| Cross-section vs time series      | <b>Cross-sectional:</b> stocks ranked against each other each month                                                                                                                       |
| Horizon and turnover              | One month; 161% of the two legs replaced monthly                                                                                                                                          |
| Market exposure                   | L/S beta -0.14 to 0.14                                                                                                                                                                    |
| Payoff shape                      | Treynor–Mazuy $\gamma$ -0.31 to 1.74 ( $t$ -0.5 to 1.7); worst 10% of market months -2.0% to +0.2% a month, best 10% -1.9% to +2.0%                                                       |
| Nearest factor theme              | investment (4 of 6 universes), HML (2 of 6 universes)                                                                                                                                     |
| Nearest library signals (returns) | Size (4 universes), 55-day breakout (3 universes), MIN5 (2 universes)                                                                                                                     |
| Economic rationale                | Recurring demand and information flows tied to the calendar (earnings seasons, dividend timing, fund flows); Keloharju et al. (2016) argue it reflects seasonal variation in risk premia. |
| Publication                       | Heston & Sadka 2008; Keloharju, Linnainmaa & Nyberg 2016                                                                                                                                  |
| Where it fits                     | In the <b>momentum</b> cluster of the library; with some information beyond its neighbours.                                                                                               |

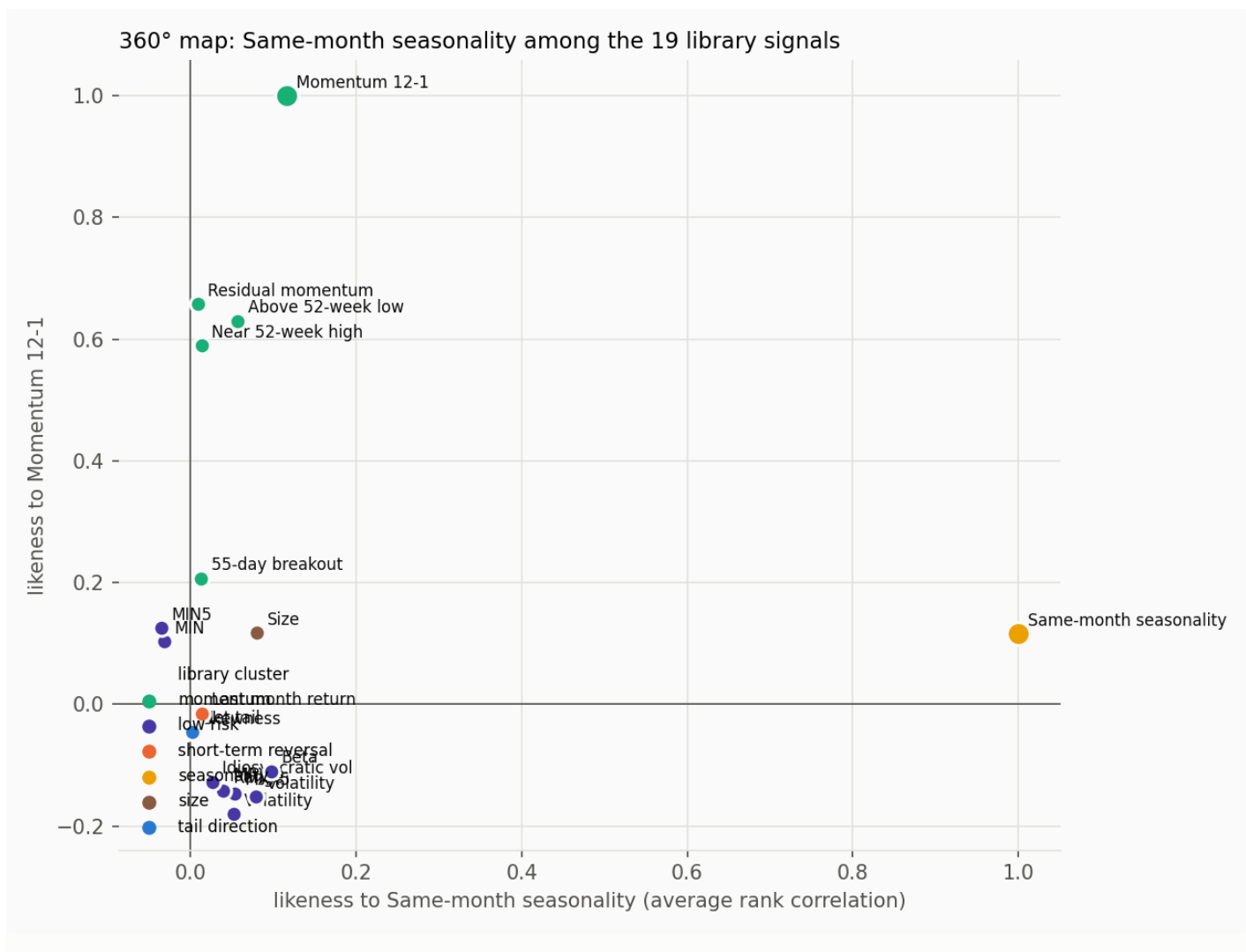


## D. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Momentum 12-1**.

| Signal (top minus bottom, raw sign)    | Cluster             | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha t ≥ +2 / ≤ -2 |
|----------------------------------------|---------------------|------------------|-------------------|-------------------|------------------|-------------------------------|
| Same-month seasonality (this strategy) | seasonality         | +1.00            | +1.00             | -1.9%             | -9.4% to +6.6%   | 0 / 1                         |
| Last month return                      | short-term reversal | +0.01            | +0.15             | +1.8%             | -3.2% to +11.0%  | 2 / 0                         |
| 55-day breakout                        | momentum            | +0.01            | +0.14             | +1.4%             | +4.1% to +13.0%  | 4 / 0                         |
| Momentum 12-1 (mirror)                 | momentum            | +0.12            | +0.14             | +7.2%             | +4.3% to +19.4%  | 5 / 0                         |
| Net tail (MAX+MIN)                     | tail direction      | +0.00            | +0.12             | -0.9%             | -5.2% to +5.2%   | 0 / 0                         |
| MIN5                                   | low risk            | -0.04            | +0.12             | -0.9%             | +2.3% to +13.7%  | 4 / 0                         |
| Near 52-week high                      | momentum            | +0.01            | +0.11             | +3.7%             | +9.3% to +22.3%  | 6 / 0                         |

| Signal (top minus bottom, raw sign) | Cluster        | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha t ≥ +2 / ≤ -2 |
|-------------------------------------|----------------|------------------|-------------------|-------------------|------------------|-------------------------------|
| Volatility (252 days)               | low risk       | +0.08            | -0.10             | +2.4%             | -12.9% to -4.8%  | 0 / 3                         |
| Idiosyncratic vol                   | low risk       | +0.03            | -0.10             | -3.6%             | -15.5% to -4.8%  | 0 / 4                         |
| MIN (worst day)                     | low risk       | -0.03            | +0.10             | -0.7%             | +3.3% to +11.1%  | 4 / 0                         |
| Beta                                | low risk       | +0.10            | -0.09             | +4.4%             | -12.6% to -4.2%  | 0 / 4                         |
| Volatility                          | low risk       | +0.05            | -0.09             | +1.1%             | -13.7% to -7.9%  | 0 / 5                         |
| Range (MAX–MIN)                     | low risk       | +0.04            | -0.08             | -0.7%             | -15.4% to -4.8%  | 0 / 4                         |
| Size                                | size           | +0.08            | +0.08             | +0.1%             | -3.8% to +9.9%   | 1 / 0                         |
| MAX5                                | low risk       | +0.05            | -0.07             | +0.6%             | -11.3% to -3.5%  | 0 / 2                         |
| Skewness                            | tail direction | +0.00            | +0.04             | -0.4%             | -6.7% to +3.9%   | 0 / 2                         |
| Residual momentum                   | momentum       | +0.01            | -0.03             | +6.4%             | +8.0% to +12.2%  | 6 / 0                         |
| MAX (best day)                      | low risk       | +0.04            | -0.02             | +0.1%             | -13.3% to -2.0%  | 0 / 2                         |
| Above 52-week low                   | momentum       | +0.06            | +0.01             | +7.5%             | +5.2% to +13.6%  | 4 / 0                         |



**Spanning.** Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t     | R <sup>2</sup> | Neighbours with $ t  \geq 2$                  |
|----------|-----------------------------|-------|----------------|-----------------------------------------------|
| US       | -7.0%                       | -2.28 | 0.09           |                                               |
| EU       | -7.8%                       | -2.10 | 0.12           | Size +0.23 (+2.8)                             |
| UK       | +3.0%                       | 0.74  | 0.14           | Size -0.25 (-2.7), Momentum 12-1 +0.18 (+2.5) |
| DK       | -8.2%                       | -2.05 | 0.03           |                                               |
| SCANDI   | -2.0%                       | -0.48 | 0.05           |                                               |
| World    | +0.6%                       | 0.25  | 0.12           | Size +0.32 (+2.7)                             |

3×3 sort, next-month return a year  
(US, EU, UK, SCANDI, World averaged)

|                               |                   |        |                    |
|-------------------------------|-------------------|--------|--------------------|
| low Same-month seasonality -  | 9.8%              | 12.2%  | 13.9%              |
| middle -                      | 9.3%              | 11.4%  | 12.8%              |
| high Same-month seasonality - | 9.7%              | 10.8%  | 13.6%              |
|                               | low Momentum 12-1 | middle | high Momentum 12-1 |

Holding Momentum 12-1 fixed (down a column), moving from low to high Same-month seasonality changes the return by -0.0, -1.4 and -0.3 points a year; holding Same-month seasonality fixed (along a row), moving from low to high Momentum 12-1 changes it by +4.2, +3.4 and +3.9. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -1% vs -2% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages -1.9%, -1.7%, +0.9%, -0.6% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)

